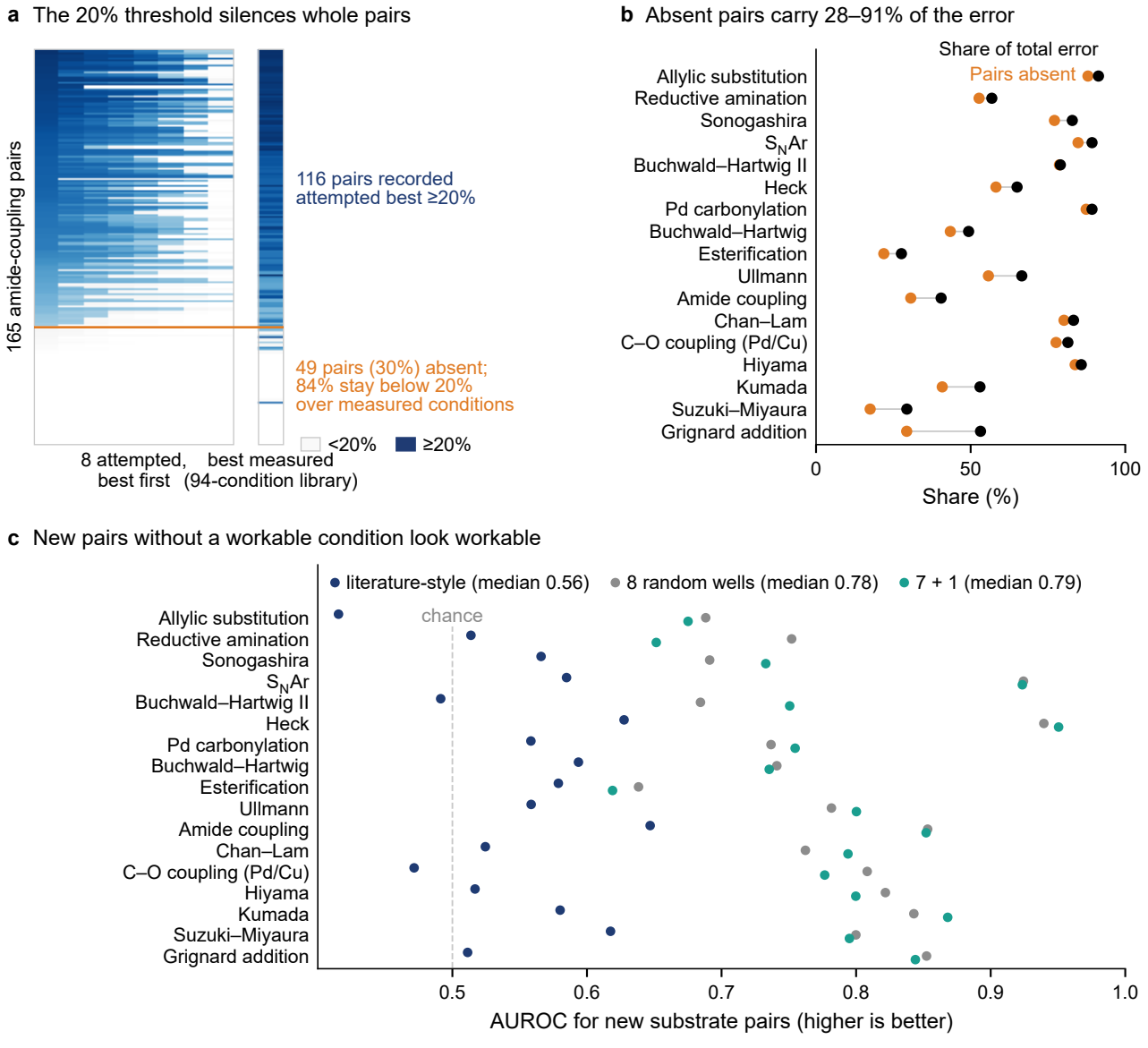
a The 20% threshold silences whole pairs
165 amide-coupling pairs
116 pairs recorded attempted best ≥20%
49 pairs (30%) absent; 84% stay below 20% over measured conditions
<20%
≥20%
8 attempted, best first
best measured (94-condition library)
b Absent pairs carry 28–91% of the error
Share of total error
Pairs absent
Allylic substitution
Reductive amination
Sonogashira
SNAr
Buchwald–Hartwig II
Heck
Pd carbonylation
Buchwald–Hartwig
Esterification
Ullmann
Amide coupling
Chan–Lam
C–O coupling (Pd/Cu)
Hiyama
Kumada
Suzuki–Miyaura
Grignard addition
0
50
100
Share (%)
c New pairs without a workable condition look workable
literature-style (median 0.56)
8 random wells (median 0.78)
7 + 1 (median 0.79)
chance
0.5
0.6
0.7
0.8
0.9
1.0
AUROC for new substrate pairs (higher is better)